HCA: Heavily Compressed Attention

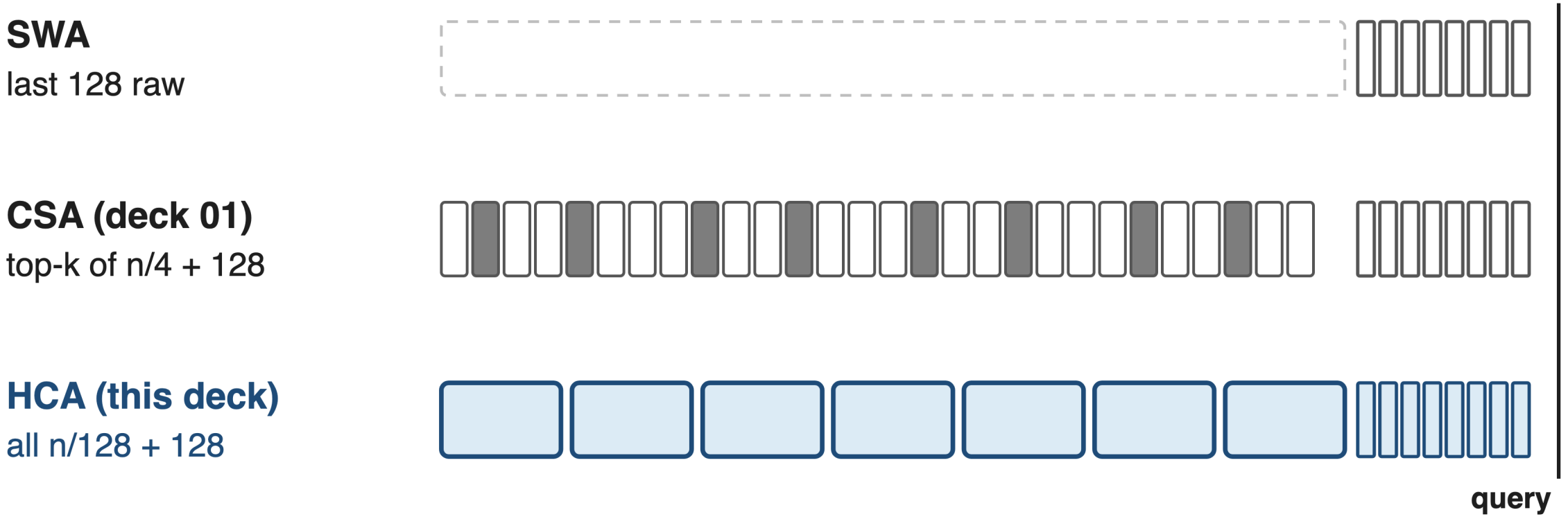
One entry per 128 tokens, interleaved with CSA

DeepSeek-AI · DeepSeek-V4

Explainer deck · Industrial Pretraining Architecture & Infra Tech Landscape (2026)

Attention & Long Context · arxiv.org/abs/2606.19348

V4 layers see context at three resolutions



HCA is 31 of 61 V4-Pro layers and 20 of 43 Flash layers; CSA, the selective type, is deck 01

A 1M-token request outgrows per-token caches

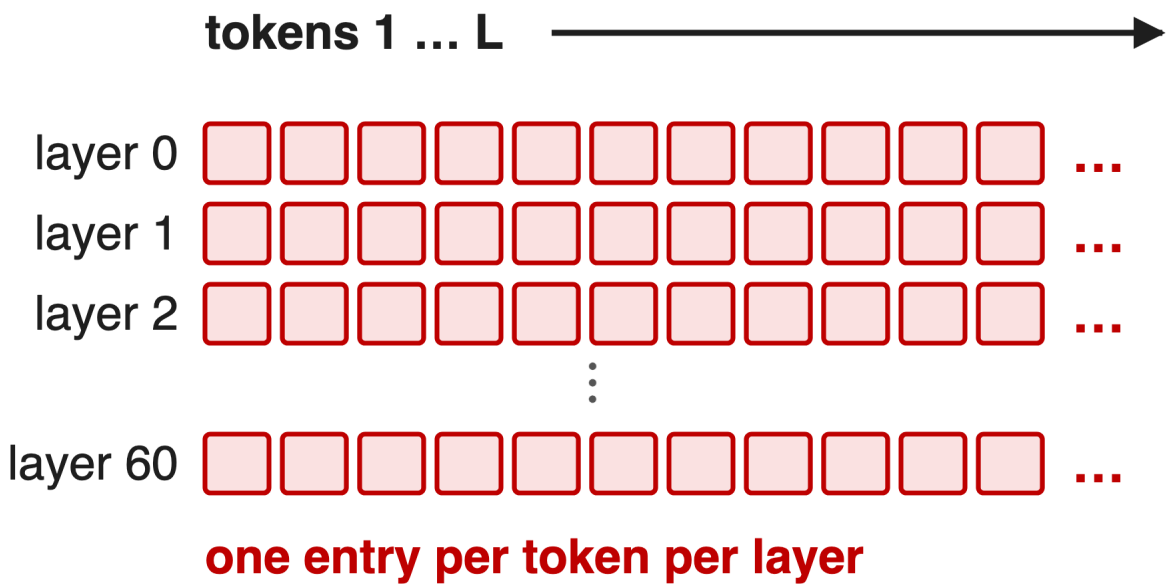
262 GB

of KV cache for one 1M-token request with BF16 GQA8 at 61 layers

Analytic: 8 KV heads × 128 dims × (K, V) × 2 B = 4,096 B per token per layer, × 61 layers × 2^{20} tokens; BF16 GQA8 is the paper's own baseline (§2.3.4)

Narrower entries still grow with every token

- **GQA / MLA**: shrink each entry, not the count
- **CSA everywhere**: count ÷ 4, still linear
- **Window only**: constant, but forgets far context



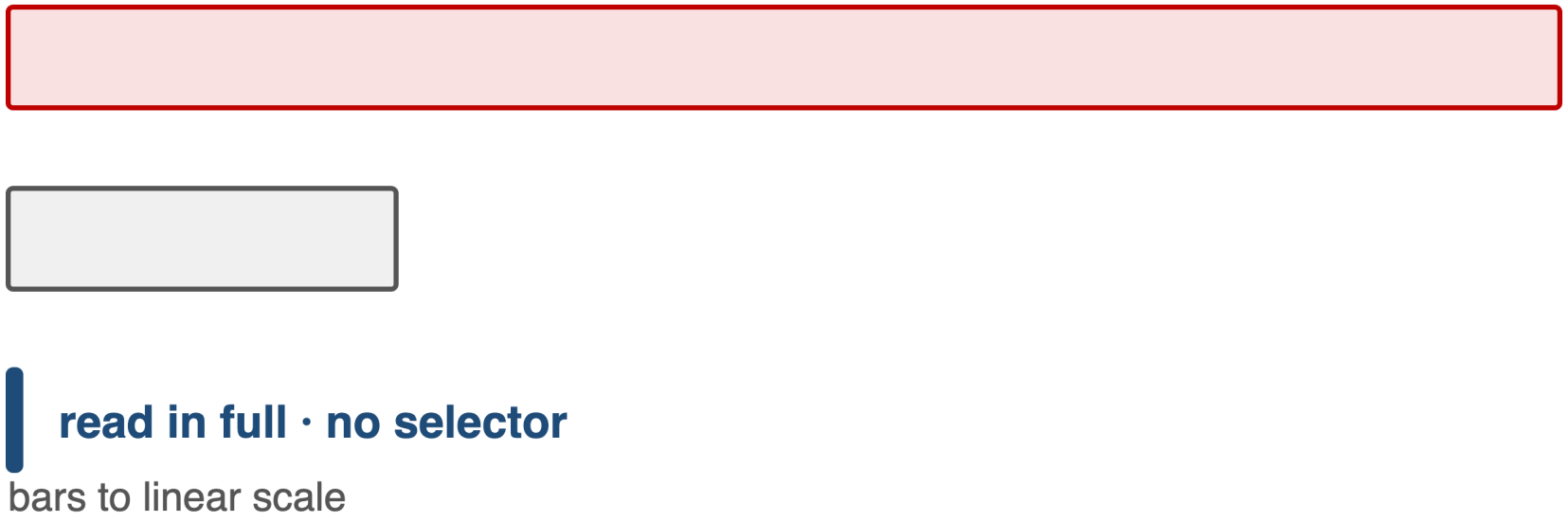
Villain: uniform resolution. Every layer stores the whole context at the same granularity

Key idea: some layers need only the gist

Raw tokens
1,048,576 per layer

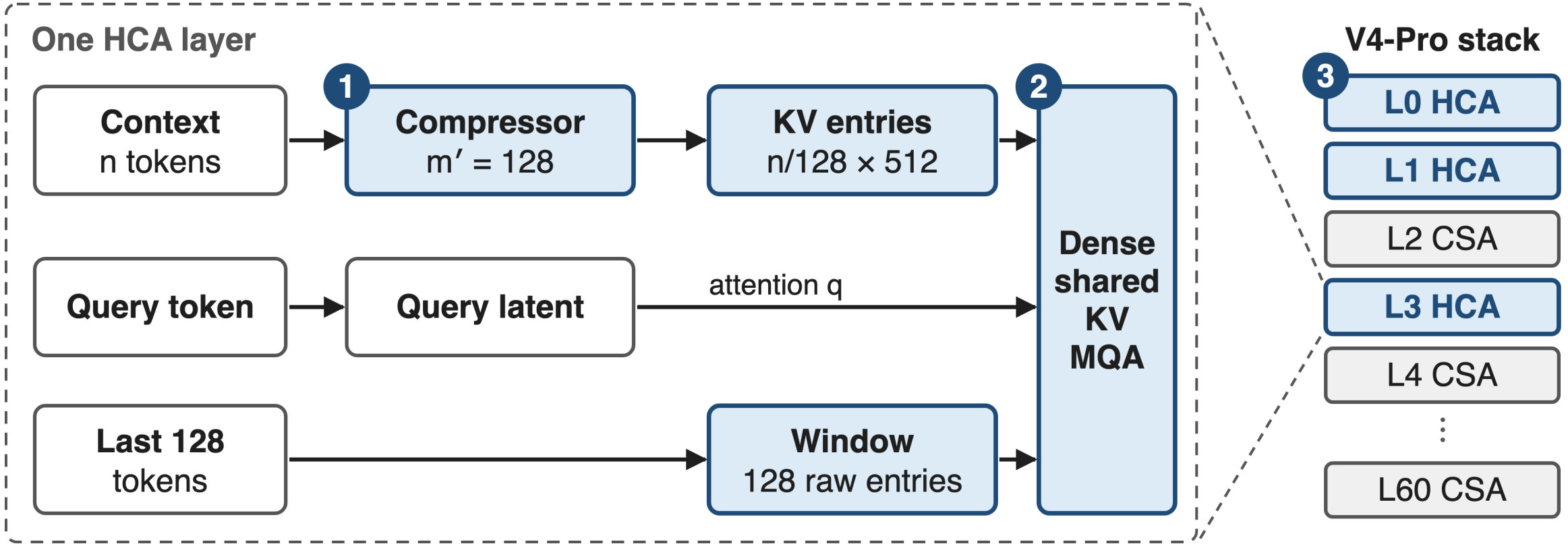
CSA entries
262,144 per layer

HCA entries
8,192 per layer



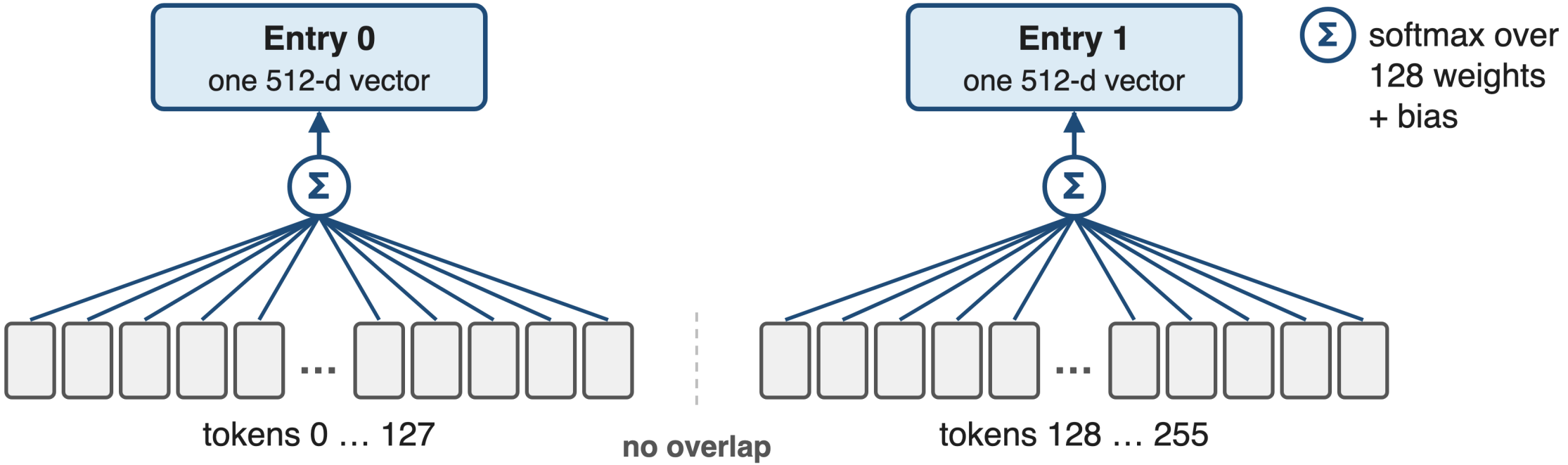
At m′ = 128, a 1M-token context is 8,192 entries: few enough to read all, no selector needed

HCA compresses 128×, reads all, and interleaves



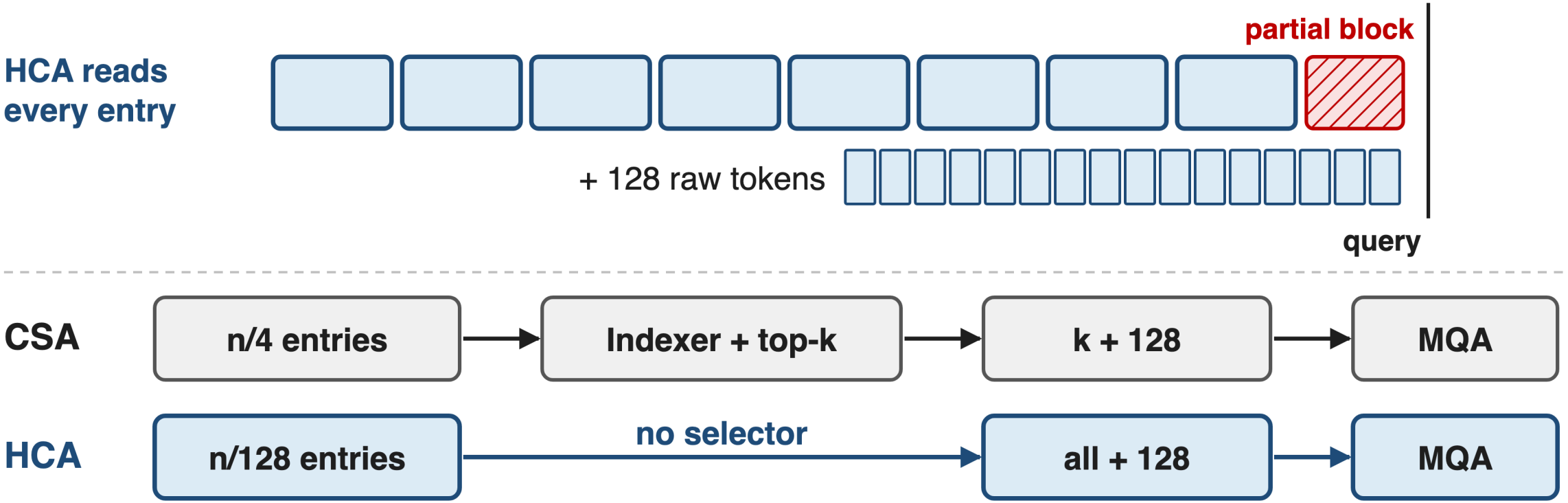
❶ pool 128 tokens → 1 entry · ❷ dense shared-KV MQA over all entries + 128 raw tokens · ❸ alternate with CSA layers

❶ Softmax pooling folds 128 tokens into one



n tokens → n/128 entries of 512 dims; unlike CSA, no overlap and no indexer keys

❷ No selector: read every entry plus a window

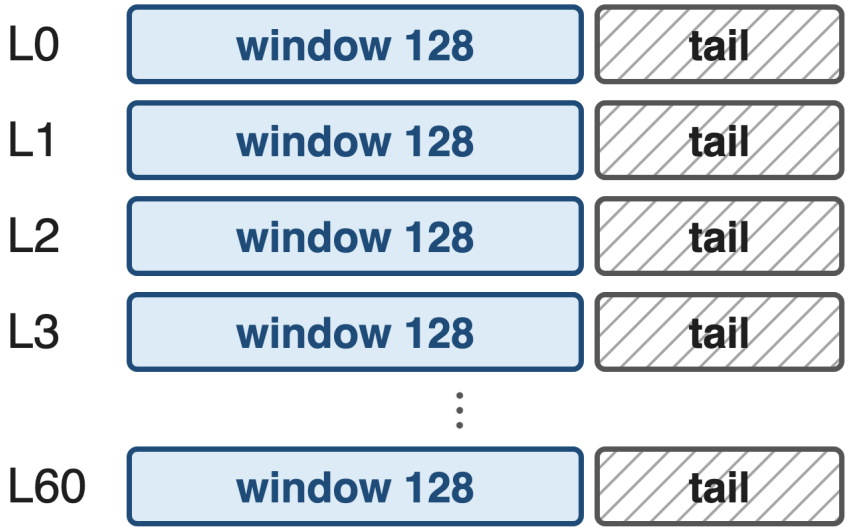


At 1M: 8,192 + 128 entries per query, 1.1 G multiply-adds per layer (analytic) vs 2.3 G for CSA

❸ Interleaving needs a two-part KV cache

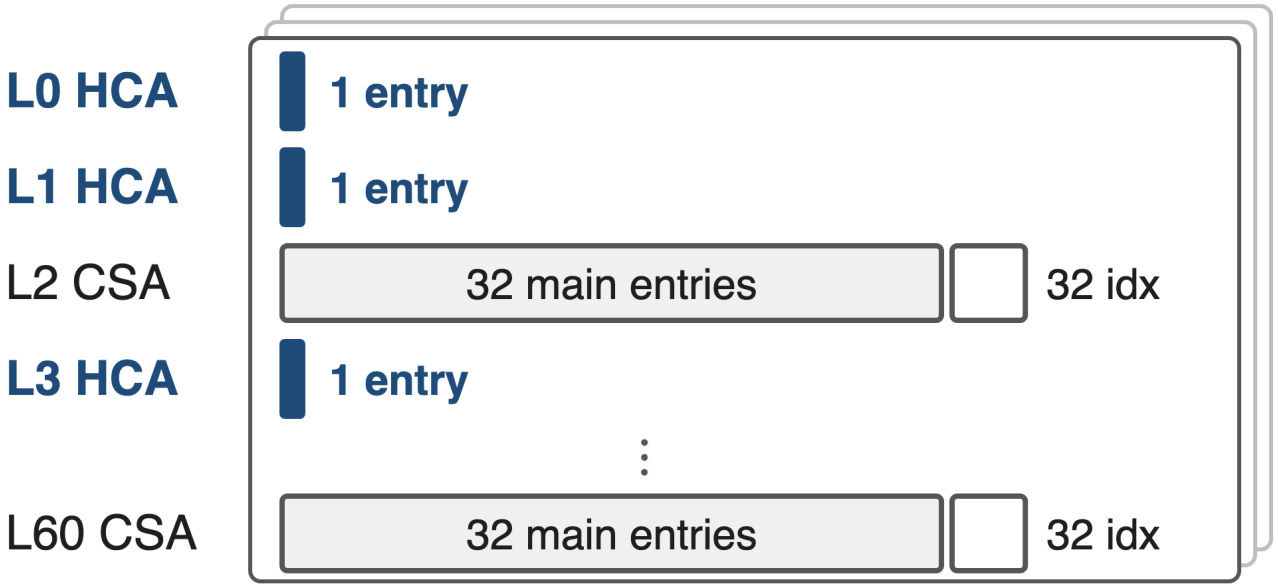
State cache

fixed size per request



Block cache

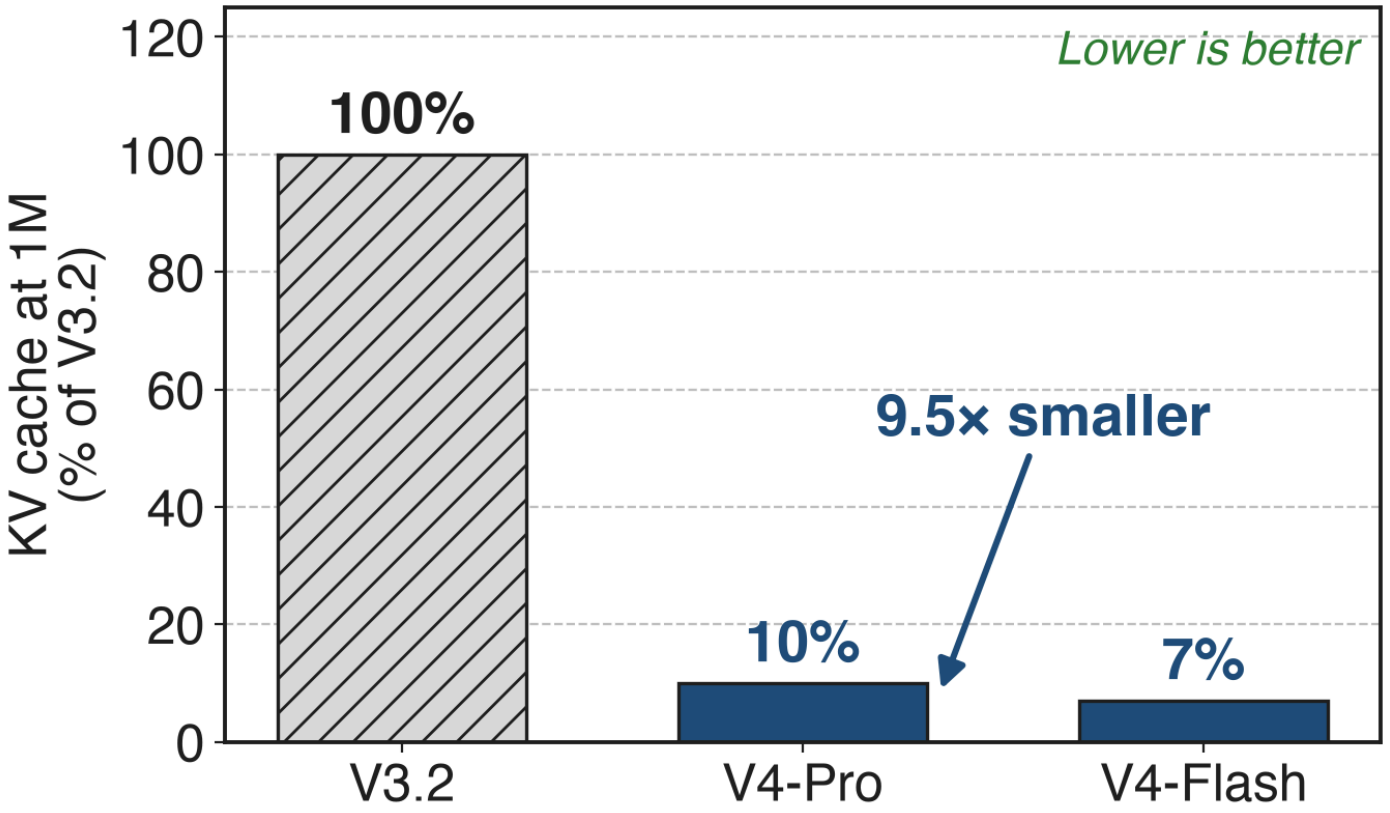
1 block = 128 tokens · N blocks grow with context



Window and not-yet-compressed tails sit in a fixed state cache; each 128-token block holds 32 CSA entries or 1 HCA entry per layer

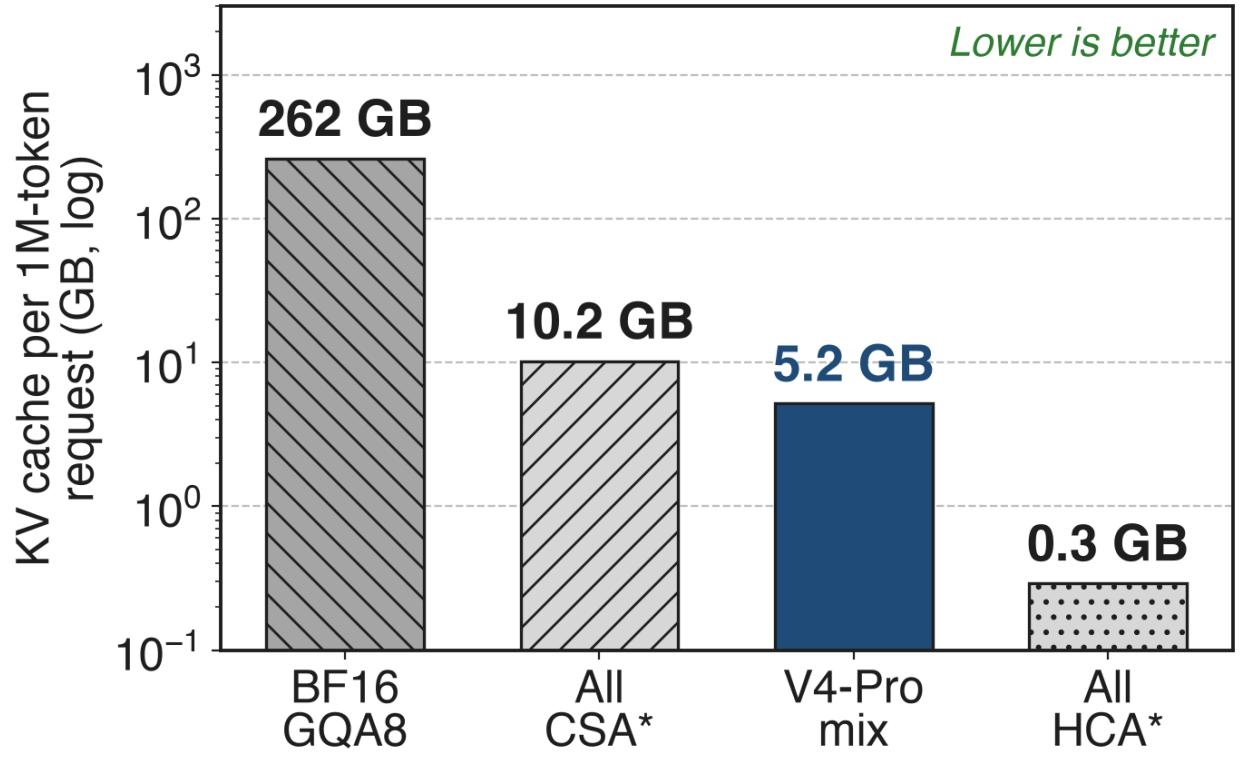
V4's 1M-token KV cache is 10% of V3.2's

- Whole stack: CSA + HCA + FP8/BF16 storage
- Flash: 7% (13.7× smaller)
- FLOPs fall too: 27% / 10% (deck 01)



The mix lands at 2% of a GQA8 cache

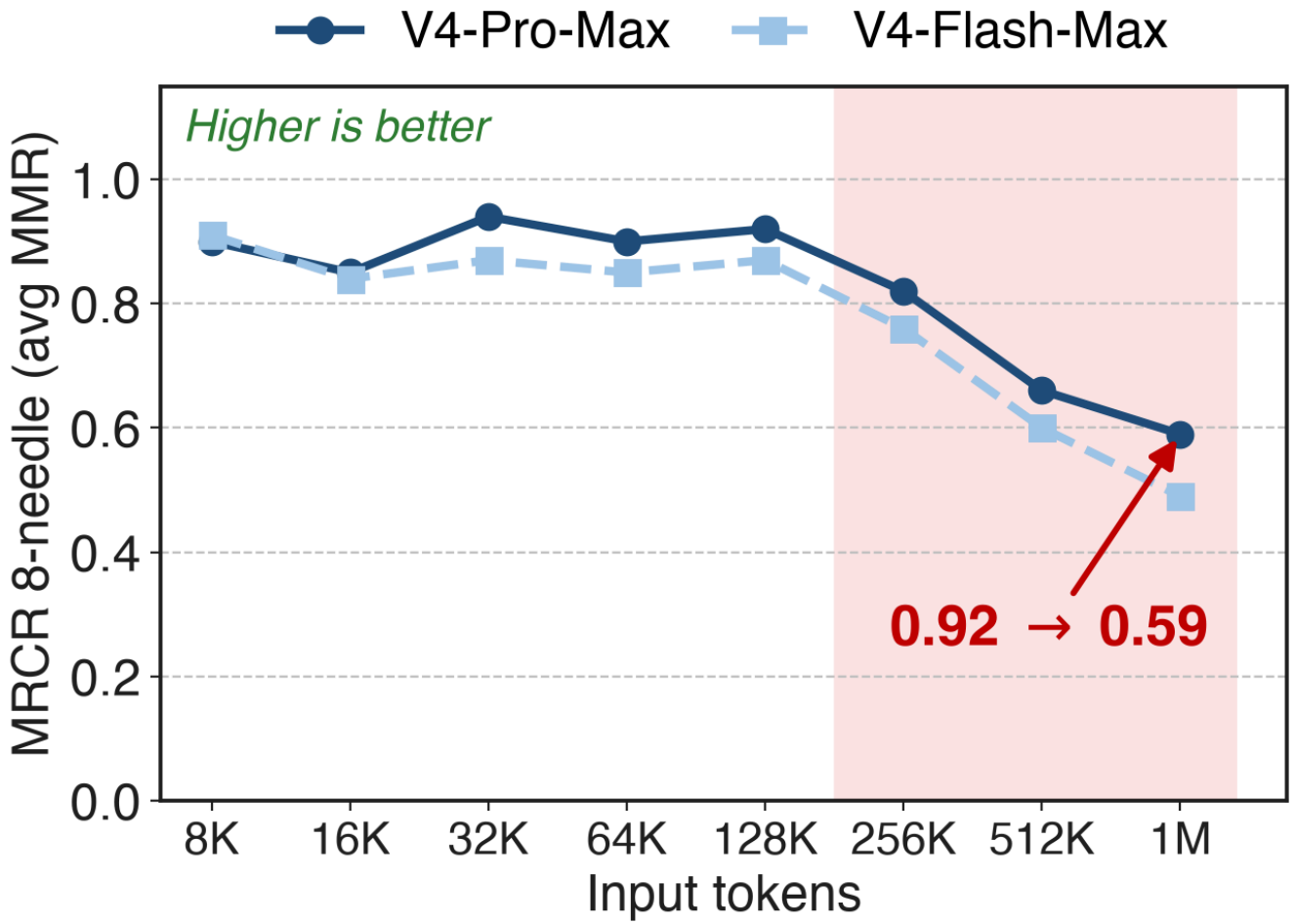
- Paper: ≈2% of BF16 GQA8 at 1M
- Our byte model: 1.98% (analytic)
- CSA layers hold 97% of it



*** Hypothetical uniform stacks: making half the layers HCA halves an all-CSA cache**

Retrieval holds to 128K, then fades

- Pro-Max: 0.92 at 128K, 0.59 at 1M
- Still ahead at 1M: 83.5 vs Gemini-3.1-Pro 76.3
- Claude Opus 4.6 leads: 92.9



Cost: coarse memory and custom cache plumbing

- Far context lives in 128-token gists; recall fades past 128K
- Custom cache: state cache, 128-token blocks, CP boundary exchange
- Disk prefix cache: window KV ≈ 8× the compressed KV

No ablation of the CSA:HCA mix or of $m' = 128$ is published

Summary

- **Pain**: 262 GB KV per 1M-token request with BF16 GQA8 (analytic)
- **Insight**: some layers need only the gist, read in full
- ❶ 128× pooling ❷ dense MQA + window ❸ CSA/HCA mix, two-part cache
- V4 at 1M: KV **10%** / **7%** of V3.2; ≈**2%** of GQA8

https://arxiv.org/abs/2606.19348